What Do Hidden Markov Models Learn From Financial Returns? Regime Detection, Redundancy, and Why Conditioning Does Not Help

Nicholas Reichert
Northeastern University
Boston, Massachusetts, USA
reichert.n@northeastern.edu

Abstract

Hidden Markov models are a standard tool for detecting latent market regimes, and conditioning a forecaster on the inferred regime is widely reported to improve prediction, particularly under stress. We re-examine that claim for equity volatility forecasting under a deliberately conservative protocol: regimes are inferred by strict online filtering with no lookahead, the train/test boundary is purged by the forecast horizon, the number of regimes and the gating rule are selected on an inner validation block rather than on the test data, and the regime model is compared against properly specified volatility benchmarks (HAR, GARCH(1,1), EWMA) rather than against a naive mean.

Under this protocol, regime conditioning does not improve volatility forecasts. Across 20 assets spanning equity indices, sectors, single names, rates, credit, commodities, FX and crypto, a pooled ridge regression on identical features beats the regime-conditioned mixture in 58 of 60 asset-horizon pairs (median −2.3% RMSE), significantly so in 37; the regime model is significantly better than the pooled ridge in none of them, and is never significantly better than HAR, GARCH(1,1) or EWMA under either squared-error or QLIKE loss.

We decompose the failure by testing every route the regime signal can take into the model. Injecting the posteriors as features, with no partitioning at all, costs 1.16 points of RMSE; gating on them costs a further 1.59; shrinking the gated mixture toward the pooled fit does not recover the difference. Correct regime timing is worth a small but real +0.69 points relative to a time-shuffled gate (Wilcoxon $p = 0.023$), which is not enough to offset either cost. Across the four variants and 240 Diebold–Mariano tests, not one shows a significant improvement.

The explanation is redundancy rather than absence of signal. The fitted high-volatility state is a good stress classifier—16% of the sample, 59% of all volatility spikes, a 3.6× lift—but it is also recoverable with 95.7% accuracy from the three HAR components alone, against a 46.2% base rate. It is a noisy, three-level, annually-refitted re-encoding of a quantity every model already reads continuously and without estimation error.

A simulation study on data whose answer we know shows the problem is not our implementation, and locates it in the estimator. Regime conditioning helps where regimes genuinely exist, so the pipeline works; real returns behave like a GARCH process with no discrete states at all. Two diagnostics that applied work routinely reports turn out not to identify regimes: state separation is as large in a world with no regimes as in one with them, and the reported persistence of the transition matrix is mostly an artefact of including an overlapping rolling-volatility feature in the emission vector—fitted to i.i.d. noise, the HMM reports substantial persistence that vanishes exactly when that feature is removed. Most strikingly, Baum–Welch identifies states by their emission distribution, so it finds states that differ in volatility *level*—which trailing volatility already reveals—and largely misses states that differ in volatility *dynamics*, which are the ones that would carry incremental information.

Three further checks agree. Gradient-boosted experts do not rescue the method. Under a Model Confidence Set the regime mixture is retained in 28% of asset-horizon pairs against 78% for GARCH(1,1). Under a decision-relevant loss—volatility targeting—the regime models are statistically indistinguishable from the pooled ridge rather than behind it, but GARCH and HAR are significantly ahead of both. We report equivalence bounds rather than bare p-values: in the median pair the data exclude improvements of *any* size at 95% confidence. Finally, the spread induced purely by the HMM's random initialization is of the same order as the effect being measured, so single-seed comparisons are uninformative.

We then quantify how a positive result can be manufactured, using this project's own first implementation as the starting point. That implementation combined four choices—a multi-horizon target that is really a single-day target at longer lead, an unstandardized fixed-penalty baseline, no train/test embargo, and selection of the regime hyperparameters on the test score—and under it the regime model shows a statistically significant improvement over its baseline. Removing the four one at a time drives that improvement to zero and then negative; correcting the baseline alone is enough to flip the sign. We release the full pipeline and propose a short checklist for evaluating regime-switching forecasters.

Keywords

volatility forecasting, regime switching, hidden Markov models, evaluation protocol, data leakage, reproducibility

1 Introduction

Financial time series are not stationary. Volatility clusters, correlations change sign in crises, and the relationship between predictors and returns appears to shift over time. This motivates *regime-switching* models, in which a latent discrete state governs the data-generating process [2, 12]. The natural machine-learning instantiation is a two-stage pipeline: infer a latent state with a hidden Markov model [27], then condition a predictor on that state, either by hard switching between per-regime experts or by soft probability weighting.

The appeal is obvious and the reported results are usually positive, and the broader turn toward machine learning in asset pricing [10] has made such two-stage pipelines easy to assemble. Our starting point was the same: an implementation of exactly this pipeline that, on its first evaluation, showed a clear improvement over a ridge-regression baseline in forecasting SPY volatility. This paper is a report of what happened when we tried to make that result robust.

The improvement did not survive. It disappeared once the linear baseline was given the same care as the proposed model, and it reversed once the target, the train/test boundary, and the model-selection step were corrected. What remains is a negative result, but a decomposable one: we can say precisely how much of the original effect each methodological shortcut contributed.

Contributions.

(1) **A causal evaluation, replicated across 20 assets.** We evaluate HMM regime-conditioned volatility forecasting under strict online filtering, an embargoed walk-forward split, nested hyperparameter selection, and a benchmark set that includes HAR [6], GARCH(1,1) [4], and EWMA [17]. Regime conditioning provides no significant gain at any horizon, for any asset, under either MSE or QLIKE [25].
(2) **A decomposition, not just a null.** We perturb the training partition and the prediction-time gate independently, and additionally test the two ways of using the regime signal that avoid gating entirely. This turns the null into an accounting identity: the regime signal costs 1.16 points of RMSE before any gating, gating adds 1.59 more, and correct timing returns only 0.69. Every route loses, and we can say by how much and why.
(3) **A mechanism.** The inferred state is 95.7% recoverable from the three HAR components alone. It is a lossy re-encoding of information the model already has, which is why no method of injecting it helps.
(4) **A simulation study identifying what the estimator can and cannot see.** Running the same pipeline on data whose generating process we control shows that (i) the standard descriptive diagnostics do not distinguish a world with regimes from one without, (ii) reported transition persistence is largely manufactured by overlapping-window emissions and survives on i.i.d. noise, and (iii) maximising emission likelihood recovers states that differ in volatility level while largely missing states that differ in volatility dynamics—that is, it finds the regimes least worth conditioning on and misses the ones that would help.
(5) **Equivalence bounds.** We report what size of improvement the data exclude, not merely that we failed to reject. In the median asset-horizon pair, no improvement of any size is consistent with the evidence at 95%.
(6) **Robustness on three further axes.** The null survives gradient-boosted experts, a Model Confidence Set over all competitors simultaneously, and a decision-relevant loss. Under the last of these the regime models are indistinguishable from the pooled ridge rather than behind it—the one place the choice of loss function matters—but the regime-*free* benchmarks are significantly ahead of both.
(7) **A protocol ablation.** Taking our own first implementation as the starting point, we quantify how much apparent improvement each of four evaluation shortcuts produces, and show that they compound: together they turn a −4% result into a statistically significant +2% one.
(8) **Seed sensitivity.** We show that the spread across HMM random initializations alone is comparable to the reported effect size, which we believe is under-appreciated in applied regime-switching work.
(9) **A reproducible pipeline and a checklist** for future work in this area.

We stress what we do *not* claim. We do not claim that market regimes do not exist, nor that regime-switching models are useless in general—they have well-documented value in asset allocation and risk management [11]. We claim that for this task, this asset, and this model class, the reported forecasting gains are an artifact of evaluation, and that the diagnostics we propose are cheap enough that they should be standard.

2 Related Work

Regime switching. Hamilton's Markov-switching autoregression [12] established the econometric template. Subsequent work applies regime switching to asset allocation [11] and documents its role in explaining return predictability [2, 28]. Most of this literature is in-sample or uses smoothed (two-sided) state estimates; our concern is the strictly causal, out-of-sample forecasting case.

Volatility forecasting. The benchmark set matters enormously here. GARCH [4, 8] remains hard to beat [13], and the HAR model [6] is the standard reference for realized volatility, whose long-memory behavior it captures with three lagged components. Andersen and Bollerslev [1] showed that apparent poor performance of volatility models was largely an artifact of using squared daily returns as the evaluation proxy; Patton [25] characterized which loss functions remain robust to such proxy noise, motivating our use of both MSE and QLIKE.

Evaluation and leakage. Our methodological concerns are not new. López de Prado [21] introduced purging and embargoing for financial cross-validation. Bailey et al. [3] formalized backtest overfitting, and Harvey et al. [15] argued for higher significance thresholds given the volume of tested strategies. Kaufman et al. [19] give the general taxonomy of leakage into which our embargo and nested-selection corrections fall. Welch and Goyal [29] performed the canonical re-evaluation of equity-premium predictors and found most did not survive out-of-sample. Outside finance, Ferrari Dacrema et al. [9] showed that weak baselines account for most reported progress in neural recommendation, Makridakis et al. [22] made the analogous point for ML forecasting, and Kapoor and Narayanan [18] document leakage as a cross-disciplinary reproducibility problem. Our contribution is to run this style of audit on the specific, popular case of HMM-based regime-conditioned forecasting, and to attribute the effect to individual protocol choices rather than merely reporting a null.

3 Problem Setup

3.1 Data

We use daily SPY (S&P 500 ETF) OHLCV data from February 2005 to July 2026, obtained from Yahoo Finance and adjusted for splits and dividends. Let p_t be the adjusted close and $r_t = \log p_t - \log p_{t-1}$ the daily log return. After feature construction and the removal of warm-up and trailing rows the sample contains 5,391 trading days, of which 3,771 (February 2011 to February 2026) are used for out-of-sample evaluation.

3.2 Targets

The forecasting target is realized volatility over the next h days,

$$y_t^{(h)} \;=\; \mathrm{RV}_{t+1:t+h} \;=\; \sqrt{\frac{1}{h}\sum_{i=1}^{h} r_{t+i}^2}, \qquad h \in \{1, 5, 20\}. \tag{1}$$

For $h = 1$ this reduces to $|r_{t+1}|$, so the family nests the single-day absolute-return target.

This definition deserves emphasis because it is a place where an error is easy to make and hard to see. A superficially similar definition, $\tilde{y}_t^{(h)} = |r_{t+h}|$, is the absolute return of a *single* day h steps ahead. It is not a multi-horizon target at all: $\tilde{y}^{(1)}$, $\tilde{y}^{(5)}$ and $\tilde{y}^{(20)}$ are the same series at different lags. The diagnostic is immediate—their sample standard deviations are 0.009186, 0.009185 and 0.009188, identical to four significant figures, whereas the aggregated targets in Eq. (1) give 0.009186, 0.007508 and 0.006837, declining with h as aggregation requires. Under the aliased definition, varying h varies only the lead time of an unchanged and extremely noisy prediction problem. We report results for both; the aliased variant is the first stage of the protocol ablation in Section 6.3.

3.3 Features

All features at time t use information available at t: lagged returns $r_{t-1}, \dots, r_{t-20}$; rolling means and standard deviations of returns over 5, 10 and 20 days; log volume change and rolling volume z-scores; and Wilder's RSI. We additionally include the three HAR components

$$\mathrm{RV}_t^{(w)} = \sqrt{\tfrac{1}{w}\textstyle\sum_{i=0}^{w-1} r_{t-i}^2}, \qquad w \in \{1, 5, 22\}, \tag{2}$$

so that every model in the comparison, including the regime-conditioned one, has access to the information the HAR benchmark uses. The final feature set has 35 columns.

4 Method

4.1 Causal regime inference

We fit a Gaussian HMM with K states and diagonal covariances on the three-dimensional emission vector $x_t = (r_t, |r_t|, \mathrm{RV}_t^{(20)})$, standardized using training-window statistics only. Parameters $(\pi, A, \{\mu_k, \Sigma_k\})$ are estimated by Baum–Welch on the training window.

The regime probability used at prediction time is the *filtered* posterior

$$\gamma_t(k) \;=\; \Pr(z_t = k \mid x_{1:t}), \tag{3}$$

computed by the forward recursion in log space,

$$\log \alpha_t(k) \;=\; \log p(x_t \mid z_t{=}k) + \log\sum_j \exp\big(\log \alpha_{t-1}(j) + \log A_{jk}\big), \tag{4}$$

normalized at each step. This is the point at which regime-switching pipelines most often leak. The smoothed posterior $\Pr(z_t = k \mid x_{1:T})$ returned by a standard `predict_proba` call, and the Viterbi path returned by `predict`, both condition on the *entire* sequence including the future, and using either one to form a forecast at time t is lookahead. We implement the forward filter explicitly, and the test-window filter is warm-started from the final training-window belief $\alpha_{T_{\mathrm{train}}}$ rather than reset to π, which is what a forecaster running the model in real time would have.

4.2 Label switching across folds

Baum–Welch state indices are arbitrary: nothing ties "state 0" in one fit to "state 0" in the next. Because the HMM is refit on every walk-forward window, concatenating raw state labels across folds interleaves different meanings. The symptom is easy to miss—the forecasts are unaffected, since within a fold the experts are fitted and applied consistently—but every per-regime table and every regime-shading plot is corrupted, showing apparent regime changes at fold boundaries that are pure relabelling.

We therefore sort states after each fit by the mean of their volatility emission, so that state 0 is always the calmest and state $K-1$ the most volatile. This is a relabelling only: it leaves the likelihood, the filtered posteriors, and every forecast unchanged, while making regime identities comparable across folds.

4.3 Regime-conditioned prediction

Given filtered probabilities, we fit one ridge expert per regime. Training rows are assigned to expert k by $\arg\max_k \gamma_t(k)$; regimes with fewer than 200 training rows fall back to a globally fitted expert. Predictions combine the experts either by hard switching, $\hat{y}_t = f_{\arg\max_k \gamma_t(k)}(x_t)$, or by soft weighting, $\hat{y}_t = \sum_k \gamma_t(k) f_k(x_t)$.

Each expert is a standardized ridge whose penalty is chosen by generalized cross-validation on its own training subsample. This matters for a fair comparison: feature standard deviations in our set span nearly four orders of magnitude (from 2.3×10^{-3} for a 20-day mean return to 11.4 for RSI), so a fixed penalty on unscaled inputs shrinks coefficients very unevenly. A baseline handicapped in this way is not a baseline.

4.4 Factorized ablations

A single "shuffle the regimes" control conflates two distinct questions. We therefore perturb the training partition and the prediction-time gate independently, giving the five arms in Table 1.

GATE_SHUFFLE is the sharpest of these. The experts are trained on genuine regimes, and at prediction time the model receives a probability vector drawn from the correct marginal distribution but attached to the wrong day. If knowing *which regime we are in today* matters, this arm must degrade.

We note one design trap we fell into ourselves. An earlier version of this ablation suite included a "uniform gate" arm intended to remove regime information while preserving the mixture structure. Because experts are assigned by $\arg\max$, a uniform probability

Table 1: Ablation design. "true" uses filtered probabilities; "shuffle" applies a random time permutation, preserving the marginal distribution of regime probabilities but destroying their alignment with the market; "single" puts all mass on one state.

Arm	Train gate	Test gate	Isolates
NORMAL	true	true	full model
GATE_SHUFFLE	true	shuffle	value of regime *timing*
EXPERT_SHUFFLE	shuffle	true	value of regime *experts*
BOTH_SHUFFLE	shuffle	shuffle	conventional shuffle control
SINGLE	single	single	pooled ridge (control)

vector selects state 0 for every row, so that arm collapsed to a single globally fitted expert and returned results numerically identical to the pooled-ridge control. It looked like an informative ablation and was in fact a duplicate.

5 Evaluation Protocol

Walk-forward splits. We use a rolling origin: six years of training (roughly 1,500 observations) followed by one year of testing, advanced one year at a time, giving 15 folds spanning February 2011 to February 2026. The HMM and all model parameters are refit on each training window.

Embargo. Because $y_t^{(h)}$ depends on returns through $t+h$, the final h rows of any training window have targets that extend into the test period. We purge them [21]. Without this, at $h = 20$ the model is fit on twenty observations whose labels are partly determined by the data it is about to be evaluated on.

Nested selection. The number of regimes $K \in \{2, 3, 4\}$ and the gating rule (hard or soft) are selected *within* each training window, on a trailing 25% validation block that is itself embargoed from the inner training portion. The selected configuration is then refit on the full training window and applied to the test fold. This is the difference between reporting what the method achieves and reporting the best of six out-of-sample numbers.

Metrics. We pool predictions across folds and report RMSE and QLIKE [25], QLIKE $= \frac{\sigma^2}{\hat{\sigma}^2} - \log \frac{\sigma^2}{\hat{\sigma}^2} - 1$, evaluated on variances. MSE and QLIKE are the two loss families that remain robust to noise in the volatility proxy, so a ranking that holds under both cannot be a proxy artifact. QLIKE diverges as $\hat{\sigma} \to 0$; we floor forecasts at the 1st percentile of realized values for reporting only.

Significance. We use the Diebold–Mariano test [7] with a Newey–West [24] long-run variance at lag $h - 1$, since overlapping targets make the loss differential autocorrelated by construction, and the Harvey–Leybourne–Newbold [16] small-sample correction. Because each model is compared against several alternatives we apply a Holm–Bonferroni correction within each target.

6 Results

6.1 Main comparison

Table 2 reports pooled out-of-sample performance.

Table 2: Pooled out-of-sample performance on realized-volatility targets, 15 walk-forward folds, Feb 2011–Feb 2026, n=3771. Lower is better. Best per column in bold. Markers report a Holm-corrected Diebold–Mariano test of the regime model against that baseline at the 5% level: $*$ the regime model is significantly better, † significantly worse. The regime model significantly beats only the trivial baselines, and significantly loses to the pooled ridge. QLIKE diverges for near-zero forecasts, which is why the zero and random-walk rows are orders of magnitude above the rest.

	$h = 1$		$h = 5$		$h = 20$	
Model	RMSE	QLIKE	RMSE	QLIKE	RMSE	QLIKE
Zero	0.010847*	1.39e+04	0.010850*	37.7831	0.010852*	8.3736
Train mean	0.008423*	2.7844	0.007042*	1.2120	0.006389*	0.8798
Random walk (RV)	0.009436*	148.9	0.005395	1.0204	0.005706	0.6026
EWMA / RiskMetrics	0.007263	1.8569	0.005234	0.5626	0.005061	0.5077
HAR	0.007133	1.8414	0.004990	0.5362	0.004967	0.4663
GARCH(1,1)	0.007052	**1.7780**	0.004884	**0.5159**	0.004913	0.4725
Ridge (pooled)	**0.006968**†	2.1185	**0.004719**	0.5585	**0.004867**†	**0.4642**
Regime-conditioned (HMM)	0.007225	4.7700	0.004925	0.5975	0.005035	0.4963

The regime-conditioned model is beaten by a pooled ridge on the same features at every horizon, significantly so at $h = 1$ and $h = 20$ after correcting for multiple comparisons. It is never significantly better than HAR, GARCH(1,1) or EWMA under either loss. The only models it does significantly beat are the trivial ones—the zero forecast, the training mean, and (at $h = 1$) the random walk in realized volatility. This is the whole difficulty with the original result in miniature: measured against a weak enough benchmark, the method looks convincing.

The two losses disagree about which benchmark is best, which is informative rather than awkward. GARCH(1,1) wins on QLIKE at $h = 1$ and $h = 5$, consistent with [13]; the feature-rich ridge wins on RMSE everywhere and on QLIKE at $h = 20$. QLIKE penalizes under-prediction heavily, so it favours the conditional-variance models that keep forecasts away from zero, while RMSE rewards the model fitted to minimize squared error. That the regime model loses under *both* is what makes the conclusion robust to the choice of loss.

As an implementation check, REGIME_SINGLE—the ablation arm that collapses the mixture to one expert—produces predictions bitwise identical to the pooled ridge at every horizon, confirming that the mixture machinery reduces correctly. Every other regime arm is worse than it.

6.2 Where does the regime signal go?

Table 3 applies the design of Section 4.4 and is the central diagnostic. On SPY, GATE_SHUFFLE—which hands the model regime probabilities attached to the wrong day—does not degrade RMSE relative to NORMAL, and at $h = 1$ and $h = 5$ is slightly *better*.

Taken alone this would suggest regime timing is worthless. The cross-asset evidence in Section 6.4 shows that reading is too strong, and we report it here as a caution against drawing mechanism conclusions from a single series: pooled over 60 asset-horizon pairs, correct timing does carry a small, statistically detectable benefit. SPY happens to be a case where it does not.

Table 3: Ablation of the regime signal (realized-volatility targets). Δ is RMSE relative to the SINGLE pooled-ridge control, in percent; negative would mean the arm beats the control. Shuffling the gate in time does not degrade the model: at $h = 1$ and $h = 5$ it improves it. If knowing the current regime mattered, the GATE_SHUFFLE row would have to be worse than NORMAL, and it is not.

	$h = 1$		$h = 5$		$h = 20$	
Arm	RMSE	Δ%	RMSE	Δ%	RMSE	Δ%
NORMAL (full model)	0.007225	+3.69	0.004925	+4.37	0.005035	+3.45
GATE_SHUFFLE	0.007106	+1.98	0.004844	+2.65	0.005090	+4.58
EXPERT_SHUFFLE	0.007044	+1.09	0.004785	+1.39	0.004925	+1.20
BOTH_SHUFFLE	0.007144	+2.53	0.004765	+0.97	0.004891	+0.50
SINGLE (pooled ridge)	0.006968	+0.00	0.004719	+0.00	0.004867	+0.00

Table 4: Protocol ablation. Improvement of the regime-conditioned model over its *matched* pooled-ridge baseline, as four evaluation shortcuts are removed cumulatively. Positive means the regime model looks better. p is a HAC-corrected Diebold–Mariano test against that baseline. At $h = 1$ the aliased and realized-volatility targets coincide by construction, so P3 and P4 are identical.

		$h = 1$		$h = 5$		$h = 20$	
	Correction applied	Δ%	p	Δ%	p	Δ%	p
P0	our original implementation	+1.93	<0.001	+1.25	0.001	+0.63	0.183
P1	+ tuned/standardized baseline	-1.03	0.041	-0.37	0.123	-0.23	0.368
P2	+ train/test embargo	-1.06	0.036	-0.34	0.152	-0.19	0.447
P3	+ nested selection	-3.69	0.001	-1.02	0.004	-0.76	0.014
P4	+ realized-vol target	-3.69	0.001	-4.37	0.028	-3.45	<0.001

The reading that survives both is that the regime label is largely a coarse restatement of trailing realized volatility, already in the feature set as $RV^{(1)}$, $RV^{(5)}$ and $RV^{(22)}$. A three-level discretization of a variable the model can already read continuously adds very little, and fragmenting the training data across experts to exploit it costs more than the little it adds. Section 6.4 puts numbers on both sides of that trade.

6.3 Protocol ablation

Table 4 and Figure 1 trace the improvement of the regime model over its matched pooled-ridge baseline as four methodological choices are corrected in sequence. The model is unchanged throughout; only the protocol moves.

The starting point, P0, is this project's own first implementation, written before any of the corrections in this paper were applied. We want to be explicit that it is not a reconstruction of any published study, and we make no claim that any particular paper combines all four choices. The claim is narrower and, we think, more useful: these four are individually easy to make, each survived our own code review, and together they produce a significant positive result on a method that does not work.

Under P0—aliased target, unstandardized fixed-penalty baseline, no embargo, and K and the gating rule chosen by out-of-sample score—the regime model improves on its baseline by 1.9% at $h = 1$

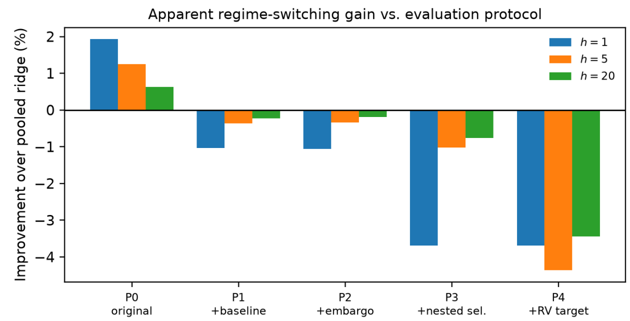

Figure 1: The apparent gain from regime conditioning as a function of the evaluation protocol, not of the model. The model is identical in every bar; only the protocol changes.

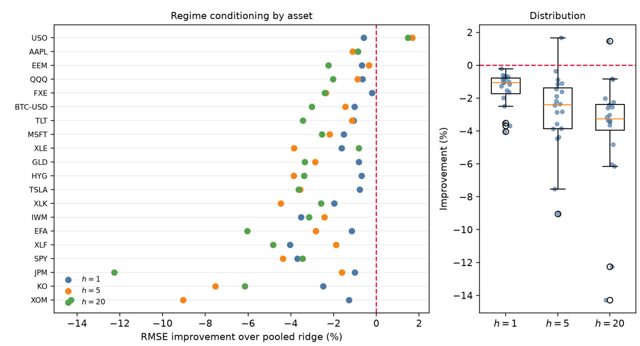

Figure 2: Cross-asset replication. Each point is one asset at one horizon; positive would favour regime conditioning. 58 of 60 cells are negative.

and 1.3% at $h = 5$, both significant at the 1% level. Correcting the baseline alone (P1) flips the sign at every horizon. Adding the embargo (P2) changes little here, which is worth saying plainly: at these horizons the leak is real but small, and we include it because its size is not knowable in advance rather than because it dominates. Nested selection (P3) removes a further 0.6 to 2.6 percentage points—the single largest correction—and the corrected target (P4) leaves the regime model between 3.4% and 4.4% worse than the pooled ridge, significantly so at every horizon.

The reason they compound is not coincidence: all four increase the flexibility of the proposed model relative to its benchmark, so their effects share a sign. That is also why no single one of them looks alarming in isolation. The literature has documented each failure mode separately—leakage across a train/test boundary [18, 21], selection reported as evaluation [3, 15], and under-specified baselines [9]—but the interaction is what turns a −4% result into a significant +2% one, and interactions are not what a checklist applied one item at a time will catch.

6.4 Does the null hold away from SPY?

A single-asset null is weak evidence, and SPY is the least favourable case one could pick: it is the most liquid and most-studied series

Table 5: Cross-asset replication. Each cell is the out-of-sample RMSE improvement of the regime-conditioned model over the pooled ridge fitted on identical features, in percent; positive would favour regime conditioning. † marks a Diebold–Mariano rejection at the 5% level, always in the direction of the pooled ridge. Assets are sorted by mean trailing volatility.

Ticker	Class	Vol	h=1	h=5	h=20
HYG	High yield credit	0.0048	-0.69	-3.87	-3.38
FXE	EUR/USD	0.0051	-0.20	-2.35†	-2.42
TLT	Long treasuries	0.0083	-1.07†	-1.14	-3.44
SPY	US large cap	0.0096	-3.69†	-4.37†	-3.45†
KO	Single name (staples)	0.0098	-2.48†	-7.53†	-6.15
GLD	Gold	0.0102	-0.81†	-2.86†	-3.35†
EFA	Developed ex-US	0.0108	-1.16	-2.83†	-6.04†
QQQ	US tech-heavy	0.0117	-0.66†	-0.88	-2.03
XLK	Technology	0.0120	-1.97†	-4.47†	-2.60†
IWM	US small cap	0.0130	-3.52†	-2.43†	-3.15†
XLF	Financials	0.0136	-4.04†	-1.88	-4.83
EEM	Emerging markets	0.0141	-0.67	-0.35	-2.25†
XOM	Single name (energy)	0.0142	-1.28†	-9.04†	-14.28†
MSFT	Single name (tech)	0.0149	-1.53†	-2.19†	-2.54†
XLE	Energy	0.0157	-1.63†	-3.85†	-0.82
JPM	Single name (financial)	0.0172	-1.00	-1.61	-12.26†
AAPL	Single name (tech)	0.0177	-0.86†	-1.10†	-0.87
USO	Crude oil	0.0201	-0.60	+1.67	+1.48
BTC–USD	Bitcoin	0.0302	-1.03	-1.45†	-3.03†
TSLA	Single name (high vol)	0.0324	-0.78†	-3.57†	-3.64†
Median			-1.05	-2.39	-3.25

available, and the one where trailing realized volatility is most informative, so it is where regime labels have least room to add anything. We therefore repeat the corrected protocol on 20 assets spanning broad equity indices, sectors, single names, long-duration rates, high-yield credit, commodities, FX and crypto, with sample histories of 12 to 21 years.

The null holds essentially everywhere. Table 5 and Figure 2 report the regime model's improvement over its matched pooled ridge: 58 of 60 asset-horizon cells are negative, with median −1.05%, −2.39% and −3.25% at $h = 1, 5, 20$. Of the 60 cells, 37 are significant deteriorations under a Diebold–Mariano test and *none* is a significant improvement. The two positive cells are both crude oil (USO) and neither is significant. Correlations between an asset's mean volatility and the size of the effect are small and inconsistent in sign (+0.14, +0.20, +0.05 across horizons), so we find no evidence that regime conditioning pays off for more volatile or less well-behaved assets.

6.5 What regime timing is actually worth

The cross-section is large enough to measure what the single-asset ablation could only bound (Figure 3). Pooling all 60 pairs, correct regime timing beats shuffled timing in 37 of them. That is not significant as a sign test ($p = 0.09$), but the paired magnitudes are: the mean benefit is +0.69 percentage points of RMSE and a Wilcoxon signed-rank test on the paired differences gives $p = 0.023$. The benefit also grows with horizon, from +0.23 points at h=1 to +0.54 at h=5 and +1.29 at h=20, which is what one would expect

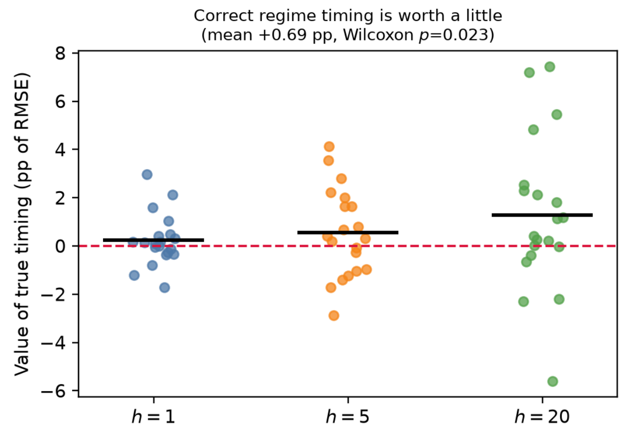

Figure 3: The paired benefit of correct regime timing over a time-shuffled gate, in percentage points of RMSE, one point per asset-horizon pair; bars are means. The benefit is small, positive on average, and grows with horizon.

Table 6: Every way of using the regime signal, against the pooled ridge, over 60 asset-horizon pairs. "Median" is the median improvement in percent (positive favours the row). "Wins" counts pairs where the row beats the pooled ridge; "sig. better" and "sig. worse" count Diebold–Mariano rejections at 5%. No regime variant achieves a single significant improvement in 240 tests, whereas GARCH and EWMA—which ignore regimes entirely—beat the pooled ridge more often than not.

Model	Median	Wins	sig. better	sig. worse
Regime gating (full mixture)	-2.30%	2/60	0	37
with time-shuffled gate	-3.14%	3/60	0	40
Regime gating, shrunk to pooled	-0.72%	11/60	0	18
Regime posteriors as features	-0.63%	4/60	0	31
HAR	-0.13%	26/60	7	8
GARCH(1,1)	+0.59%	39/60	4	3
EWMA	+0.42%	33/60	14	9

if the regime state carries information about persistent volatility levels rather than about tomorrow specifically.

So regime timing is not worthless. The question is whether it is worth enough, and whether the gating mechanism is what stands in the way.

6.6 Can the regime signal be used without gating?

If the binding constraint were the partition, then using the regime posteriors *without* partitioning should recover the loss. We test the two obvious ways to do that, both under the same protocol. The first appends the filtered posteriors $\gamma_t(1), \ldots, \gamma_t(K)$ to the feature matrix of a single pooled ridge, so the regime signal is available to the model at no partitioning cost whatsoever. The second shrinks the gated mixture toward the pooled fit by a factor selected on the

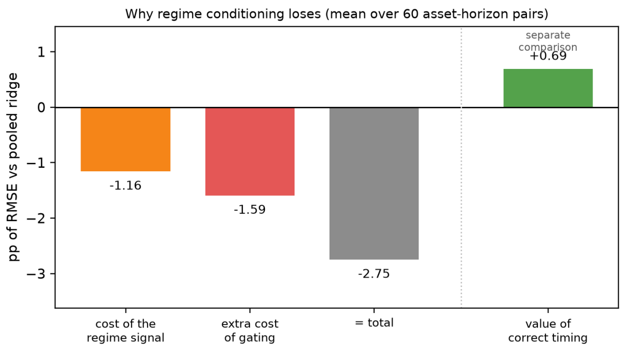

Figure 4: Decomposition of the regime model's loss against the pooled ridge. The first two bars are exactly additive by construction. The timing term is a different comparison—gated versus gated-with-shuffled-gate—and is not a component of the total.

inner validation block, with a grid that includes both endpoints, so it can recover either extreme if the data prefer them.

Neither rescues the method. Table 6 reports all four routes. Feature augmentation loses by 0.63% at the median and shrinkage by 0.72%, against 2.30% for full gating. Across the four variants and 240 Diebold–Mariano tests, *none* produces a significant improvement over the pooled ridge.

Figure 4 separates the two costs. Injecting the regime signal at all costs 1.16 points of RMSE on average (Wilcoxon $p = 2.4 \times 10^{-9}$); gating on it rather than merely conditioning on it costs a further 1.59 (paired Wilcoxon $p = 8.7 \times 10^{-8}$). Both are real, and the partition is the larger of the two—so the mechanism does matter—but removing it entirely still leaves the regime model behind a plain ridge.

Two details are worth recording. The shrinkage grid includes $\lambda = 1$, which is exactly the pooled ridge, and the inner validation block selects it in only 15% of folds, choosing a mean shrinkage of 0.45 instead. In-sample the regime signal looks useful often enough to be selected, and out-of-sample it is not. And the benchmarks that ignore regimes entirely do better than the ridge we are using as the reference: GARCH(1,1) beats it in 39 of 60 pairs and EWMA in 33. The regime model is therefore losing to a control that is not even the strongest available baseline.

6.7 What does the HMM actually learn?

A null result on real data is not interpretable on its own: it could mean markets have no regimes, or that our pipeline cannot find regimes that are there. We settle this by running the identical pipeline on synthetic data from worlds whose answer we know—two with genuine Markov switching, one with persistent continuous volatility and no discrete states, and two with no dynamics at all. The processes are calibrated to SPY's unconditional volatility, excess kurtosis and absolute-return autocorrelation, and the GARCH world uses Student-t innovations so that it is a fair stand-in for real returns rather than one distinguishable on tail thickness.

Table 7 contains three findings.

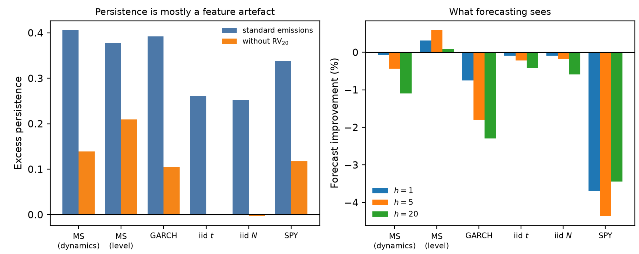

Figure 5: Left: apparent regime persistence, with the standard emission vector and with the 20-day rolling volatility removed. On i.i.d. data the first is large and the second is zero. Right: forecast improvement from conditioning on the inferred state. Only the right panel separates the worlds that contain regimes from those that do not.

The usual descriptive diagnostics do not identify regimes. State separation is 3.00 in the GARCH world, which contains no discrete states whatsoever, against 3.15 for genuine two-state switching. A reader shown only the fitted means, the transition matrix and a shading plot could not tell the two apart—and both look like the real data.

Reported persistence is largely an artefact of the emission vector. This is the sharpest result. Fitted to i.i.d. Gaussian returns—no volatility clustering, no regimes, nothing—the HMM reports excess persistence of +0.25, i.e. a self-transition probability roughly double the stationary probability. Remove the 20-day rolling volatility from the emission vector and it falls to −0.003, exactly the zero it should be. Consecutive values of $\mathrm{RV}^{(20)}$ share 19 of their 20 observations, so the filter is reading back autocorrelation the feature construction handed it. The effect is not small relative to the real signal: of SPY's apparent +0.339, only +0.118 survives the removal, and of the GARCH world's +0.392, only +0.105 does. Since "the inferred regimes are highly persistent" is among the most commonly reported diagnostics in this literature, we would encourage reporting it in the ablated form.

Only forecasting discriminates, and it places real data with GARCH.. Regime conditioning helps in the world where regimes genuinely exist and differ in level (+0.31% and +0.59% at $h = 1, 5$, both significant across 12 replications), and hurts in every world where they do not. Real SPY sits with GARCH, not with Markov switching. Our pipeline is therefore capable of detecting regimes when they are present—the positive control works—which is what licenses reading the empirical null as a statement about markets rather than about our implementation.

An unexpected asymmetry. We designed the "differing dynamics" world to be the *strongest* case for a regime-conditioned mixture: its two states share a similar volatility level but differ in how fast volatility mean-reverts, so per-regime experts have genuinely different coefficients to learn. Regime conditioning fails there (−0.44% at $h = 5$), and the state-recovery column explains why: the adjusted Rand index against the true path is 0.13, against 0.33 in the world where states differ in level.

Table 7: What the HMM learns from each world. All rows use the identical pipeline. "Separation" is the ratio of the highest to the lowest state's mean realized volatility; "excess persistence" is the average self-transition probability minus the stationary probability, so zero means states are temporally independent. Persistence is reported with the standard emission vector and again with the 20-day rolling volatility removed; on data with no dynamics at all the first is large and the second is zero, so most of what the usual diagnostic reports is autocorrelation contributed by an overlapping-window feature. "State recovery" is the adjusted Rand index against the true latent path where one exists. The final columns are the out-of-sample RMSE improvement from conditioning on the inferred state. Separation and persistence do not distinguish the worlds containing regimes from those that do not; forecasting does.

Data-generating process	True regimes?	Separation	Excess persistence std.	Excess persistence no RV_{20}	State recovery	Forecast improvement h=1	h=5	h=20
Markov switching (differing dynamics)	✓	2.80	+0.406	+0.139	0.13	-0.07%	-0.44%	-1.10%
Markov switching (differing level)	✓	3.15	+0.378	+0.209	0.33	+0.31%	+0.59%	+0.09%
GARCH(1,1)-t	×	3.00	+0.392	+0.105	–	-0.75%	-1.79%	-2.29%
i.i.d. Student-t	×	1.60	+0.261	+0.002	–	-0.09%	-0.22%	-0.42%
i.i.d. Gaussian	×	1.28	+0.253	-0.003	–	-0.09%	-0.17%	-0.59%
SPY (real data)	–	2.51	+0.339	+0.118	–	-3.69%	-4.37%	-3.45%

This is a statement about the estimator rather than about our setup. Baum–Welch maximises the likelihood of the *emissions*. It therefore identifies states that differ in their emission distribution, and is close to blind to states that differ only in how the process evolves. But a state that differs in level is one that trailing realized volatility already reveals, while a state that differs in dynamics is exactly the one that would carry incremental information. The estimator is best at finding the regimes that are least worth conditioning on, and worst at finding the ones that would help—not because it is badly implemented, but because its objective and the forecasting objective are different.

6.8 Two fixes the diagnosis implies

The two failures identified above—an emission vector that manufactures persistence, and an objective that finds the wrong states—each suggest a remedy. We test both.

Emissions without overlapping windows. Replacing the emission vector with volatility-standardised returns $z_t = r_t/\hat{\sigma}_{t-1}$, where $\hat{\sigma}$ is a causal EWMA of past squared returns, removes the manufactured persistence exactly. Excess persistence on i.i.d. Gaussian returns falls from +0.301 to −0.005, and on i.i.d. Student-t from +0.297 to +0.002. More usefully, the diagnostic now *discriminates*: the GARCH world, which has no discrete states, falls to +0.013, while the two genuine regime worlds retain +0.065 and +0.048. Under any process whose only structure is time-varying scale, the standardised series is approximately i.i.d., so surviving persistence indicates structure the volatility level does not already explain.

Applied to SPY, this returns −0.001. Real equity returns, once their own volatility level is divided out, exhibit *less* regime persistence than a GARCH process containing no regimes at all. The apparent regime structure in Figure 7 is the volatility level and nothing else. We consider this the single cleanest statement the paper can make, and it is a measurement rather than an inference.

Estimating the states jointly with the forecast. If the problem is that Baum–Welch optimises the wrong objective, the fix is to make the emission the predictive density itself, $p(y_t \mid x_t, z_t{=}k) = \mathcal{N}(\beta_k' x_t, \sigma_k^2)$, and estimate β, σ and the transition matrix jointly by EM. This is Hamilton's [12] Markov-switching regression; the estimator is not new. What is new is the controlled comparison against the two-stage pipeline, on data whose regimes we control, with both arms given the same protocol—embargoed splits, K and the ridge penalty selected on the same inner validation block, and a horizon-aware causal filter for out-of-sample prediction.

The results split three ways.

Where regimes exist and are identifiable, joint estimation is a large improvement: on the level-switching world it gains +2.62% and +3.91% at $h = 1, 5$ against the two-stage pipeline's +0.37% and +0.72%, roughly a sevenfold difference. The diagnosis is therefore correct as far as it goes: the objective mismatch is real, and removing it recovers substantially more of the available signal.

But joint estimation does *not* rescue the dynamics-switching world, which is the case we expected it to fix. It loses there at every horizon, and beats the two-stage pipeline in only 1 of 12 replications. We had predicted the opposite. The implication is stronger than the one we set out to demonstrate: states that share a volatility level and

Table 8: Two-stage versus jointly-estimated regimes. Both are RMSE improvement over the same pooled ridge, in percent, averaged over four replications for the synthetic worlds. Both arms use the identical protocol: embargoed walk-forward splits, K and the ridge penalty selected on the inner validation block, and causal prediction. Joint estimation is much better where regimes exist and differ in level, no better where they differ only in dynamics, and markedly worse where there are no regimes to find.

Data-generating process	Two-stage h=1	h=5	h=20	Joint h=1	h=5	h=20
Markov switching (level)	+0.37	+0.72	-0.66	+2.62	+3.91	-6.97
Markov switching (dynamics)	+0.13	+0.14	-0.70	-1.66	-0.98	-4.54
GARCH(1,1)-t (no regimes)	-0.98	-3.52	-6.44	-4.80	-7.38	-11.84
SPY (real data)	-1.13	-4.58	-3.56	-8.10	-11.41	-2.72

differ only in how volatility evolves appear not to be recoverable from a return series by *either* estimator. That is a statement about the information content of returns rather than about the choice of objective, and it bounds what any regime detector operating on returns alone can deliver.

Finally, where there are no regimes—the GARCH world and real data—joint estimation is markedly worse than the two-stage pipeline (−4.80% and −8.10% at $h = 1$). With K times the parameters and nothing to find, the additional flexibility only overfits. This is consistent with the emission diagnostic above: on SPY there is no residual regime structure for a better estimator to exploit, so a more powerful estimator simply does more damage.

The practical recommendation follows directly. Run the standardised-emission diagnostic first. If it returns approximately zero, as it does for SPY, no regime machinery of any sophistication will help, and the correct move is to condition on volatility continuously and stop. If it returns something materially positive, prefer joint estimation to the two-stage pipeline—but expect it to find only regimes that differ in level.

6.9 Nonlinear experts

A natural objection is that our experts are linear, and that a regime label would earn its keep only if it gated genuinely different functional forms. We therefore repeat the comparison with gradient-boosted experts, pooled and gated, holding K fixed at 3.

The objection does not survive, but not for the reason we expected. Gradient boosting is substantially *worse* than ridge at this task—a median 6.03% higher RMSE pooled, and it wins in 0 of 60 pairs—which is consistent with the long-standing observation that realized volatility is close to linear in its own lagged components, the fact HAR is built on [6]. Gating those experts leaves them at 6.29%, so the whole branch is uncompetitive regardless of how the regime signal is used.

The interesting part is the gating cost. With ridge experts, gating costs 2.30 points (Wilcoxon $p = 1.6 \times 10^{-10}$); with boosted experts it costs 0.11 points and is indistinguishable from zero ($p = 0.37$). We had predicted the opposite in an earlier draft—that more flexible experts would make the partition more expensive, since each is fitted on a fraction of the window—and that prediction is wrong. The likelier explanation is that a boosted model can already represent regime-dependent behaviour internally through its splits, so explicit gating is redundant rather than costly, and that its own variance dominates any marginal partition effect. We record the failed prediction because it bears on the paper's method: the partition cost is a property of the expert family, not a universal, and our decomposition in Section 6.6 should be read as describing linear experts specifically.

6.10 Which models are actually best?

The tests so far are pairwise. With a dozen competitors that is the wrong question: a model can fail to lose against any single rival and still be excluded once all of them are considered jointly. The Model Confidence Set [14] returns the set of models that cannot be distinguished from the best, which is the object our claim really concerns. We compute it separately for each of the 60 asset-horizon pairs at the 90% level, using a stationary block bootstrap [26] with block length h to preserve the dependence that overlapping targets induce.

Table 9: Model Confidence Set [14] at the 90% level, computed separately for each of the 60 asset-horizon pairs with a stationary block bootstrap at block length h. Entries are the percentage of pairs in which the model survives into the set of models that cannot be distinguished from the best. Regime-conditioned models are in bold. Unlike the pairwise tests, this accounts for the full set of competitors simultaneously.

Model	MSE	QLIKE
Ridge (pooled)	75%	67%
HAR	52%	65%
GARCH(1,1)	78%	88%
EWMA	77%	85%
Gradient boosting (pooled)	18%	43%
Regime gating	28%	42%
Regime as features	40%	55%
Regime gating, shrunk	60%	72%
Regime gating, GBM experts	15%	35%

Table 9 reports how often each model survives. The ordering is unambiguous. GARCH(1,1) and EWMA—neither of which knows anything about regimes—are retained in 78% and 77% of pairs under MSE and in 88% and 85% under QLIKE. The full regime mixture is retained in 28% and 42%. Boosted regime experts, the most elaborate model we fit, are retained least often of all, in 15% and 35%.

The one regime variant that holds up is the shrunk mixture, at 60% and 72%, and its explanation is deflationary: shrinkage selected on the inner split pulls it most of the way back toward the pooled ridge, so what survives is largely the ridge. Across all four regime variants and both losses, no variant is retained as often as the simplest volatility model in the comparison.

6.11 Does it pay for a decision?

Squared error is not what volatility forecasts are for. A model with worse RMSE but a 3.6× lift on stress classification could still be the better input to a risk system, and if it were, that would be this paper's one positive finding. We test two standard uses.

Volatility targeting. Position size is set to $\sigma^{\text{target}}/\hat{\sigma}_t$ for a 10% annualised target [23], and we measure how far realized volatility lands from that target. This has no distributional assumption and the loss is directly interpretable.

Value-at-Risk. The forecast is converted to a 5% quantile by filtered historical simulation on a trailing window—so the scaling uses only past information—and coverage is tested with Kupiec [20] and Christoffersen [5].

The economic evidence, in Table 10, is softer than the statistical evidence and in one respect kinder to regime conditioning. On volatility targeting, no regime variant differs significantly from the pooled ridge: the full mixture is marginally worse (paired Wilcoxon $p = 0.053$) and the shrunk mixture marginally better ($p = 0.34$), neither conclusive. So under a decision-relevant loss the regime models are merely *indistinguishable* from the pooled ridge, rather

Table 10: Economic evaluation, median over 20 assets. Volatility targeting sizes a position at $\sigma^{\text{target}}/\hat{\sigma}_t$ for a 10% annualised target; the reported error is |realised − target| annualised volatility, so lower is better. VaR is at the 5% level with the forecast-to-quantile scaling calibrated by filtered historical simulation on a trailing window, evaluated at h=1 where the realised signed return is unambiguous. "Pass CC" is the share of assets passing Christoffersen conditional coverage at 5%.

	Vol-target error			VaR (5%, h=1)	
Model	h=1	h=5	h=20	Rate	Pass CC
Ridge (pooled)	0.0375	0.0104	0.0049	0.052	95%
HAR	0.0347	0.0099	0.0040	0.052	95%
GARCH(1,1)	0.0337	0.0093	0.0031	0.052	90%
EWMA	0.0350	0.0109	0.0040	0.052	75%
Gradient boosting (pooled)	0.0389	0.0116	0.0074	0.053	65%
Regime gating	0.0370	0.0111	0.0056	0.053	95%
Regime as features	0.0369	0.0119	0.0056	0.053	95%
Regime gating, shrunk	0.0360	0.0099	0.0046	0.051	95%
Regime gating, GBM experts	0.0392	0.0130	0.0085	0.053	65%

than significantly behind it as they are under RMSE. That difference between loss functions is worth recording.

It does not, however, amount to a case for regime conditioning, because the regime-free benchmarks are significantly better than both. GARCH(1,1) reduces the volatility-targeting error relative to the pooled ridge at every horizon ($p < 10^{-4}$, better in 43 of 60 pairs) and HAR does so too ($p = 0.006$). Whatever the regime models are indistinguishable from, it is not the frontier.

The VaR results are largely uninformative about model quality, and we say so rather than presenting them as support. Because filtered historical simulation recalibrates the forecast-to-quantile scaling on a trailing window, it absorbs level errors, and every non-degenerate model attains roughly nominal coverage (5.1–5.3%) and passes conditional coverage in 85–95% of assets. The one model that fails badly—the random walk in realized volatility, passing in 10% of assets—does so because its forecasts are erratic rather than mis-levelled. The dimension that does separate models is the average VaR level at matched coverage, i.e. how much capital the forecast ties up, and there the regime mixture is the least efficient of the well-calibrated models (−0.0211 against −0.0207 for GARCH). The margin is small and we would not build an argument on it.

6.12 What the data can exclude

A failure to reject equal accuracy is not evidence of equal accuracy; it is also what an underpowered test produces. Since our central claim is a null, the burden is on us to show the study could have detected an effect had one existed.

Table 11 therefore reports, for each asset-horizon pair, the upper limit of a 95% confidence interval on the regime model's RMSE improvement over the pooled ridge—the largest improvement the data do not exclude. The interval is built on the same HAC-corrected loss differential as the Diebold–Mariano tests and mapped monotonically onto the RMSE percentage scale.

In the median pair the upper limit is itself negative at every horizon, meaning no improvement of any size is consistent with the evidence at 95%. The bounds are tightest at h=1, where even the least informative asset excludes improvements above 0.80%. They loosen at longer horizons—the weakest single case at h=20 only excludes improvements above 10.3%—because overlapping 20-day targets leave far fewer effective observations, which is precisely what the HAC correction is there to reflect. We would not claim a tight bound at h=20 for every asset; we would claim that at h=1, and in the median case at all horizons, the null is a measurement rather than an absence of one.

Table 11: Equivalence bounds. For each asset-horizon pair we form a 95% confidence interval on the regime model's RMSE improvement over the pooled ridge and report the upper limit, i.e. the largest improvement the data cannot exclude. A negative bound means *no* improvement of any size is consistent with the data at 95%. This is the statement a null result requires: failing to reject equal accuracy is also what an underpowered test produces.

Variant	Horizon	Median bound	Worst case	Bound < 0
Regime gating (full mixture)	$h = 1$	-0.12%	+0.80%	13/20
	$h = 5$	-0.47%	+6.14%	13/20
	$h = 20$	-0.11%	+10.34%	11/20
Regime posteriors as features	$h = 1$	-0.02%	+0.64%	11/20
	$h = 5$	-0.15%	+4.39%	11/20
	$h = 20$	+0.22%	+3.89%	9/20
Regime gating, shrunk to pooled	$h = 1$	+0.13%	+0.94%	6/20
	$h = 5$	+0.46%	+6.52%	7/20
	$h = 20$	+0.50%	+6.94%	5/20

Table 12: Sensitivity to the HMM's random initialization. Ten seeds, everything else fixed. Δ% is improvement over the (deterministic) pooled ridge. The spread across seeds is comparable to the effect size the comparison is trying to resolve.

Horizon	Pooled ridge	RMSE mean	RMSE s.d.	Δ% min	Δ% max
$h = 1$	0.006968	0.007114	0.000081	-3.69	-0.18
$h = 5$	0.004719	0.004948	0.000059	-6.71	-1.94
$h = 20$	0.004867	0.005043	0.000046	-5.19	-2.36

6.13 Seed sensitivity

Baum–Welch maximizes a non-convex likelihood. Table 12 and Figure 6 report the spread of out-of-sample RMSE across ten random initializations, with everything else held fixed. The range is comparable to the differences the model comparison is trying to resolve, and the number of regimes chosen by nested selection is itself seed-dependent. A single-seed comparison between a regime model and a baseline is therefore close to uninformative, yet single-seed results are the norm in this literature.

We emphasise that every seed still leaves the regime model behind the pooled ridge—the conclusion does not depend on the draw. What the spread shows is that the *magnitude* reported by any one run is not a property of the method, and that a study reporting a small positive effect from a single seed has not measured anything. We recommend reporting the seed distribution as a matter of course.

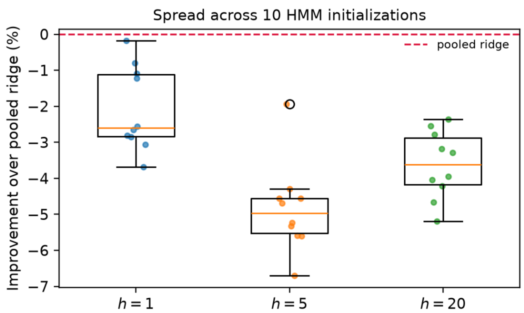

Figure 6: Improvement over the pooled ridge across ten HMM initializations. Every seed is negative, but the spread is wide enough that a single run could report anything within it.

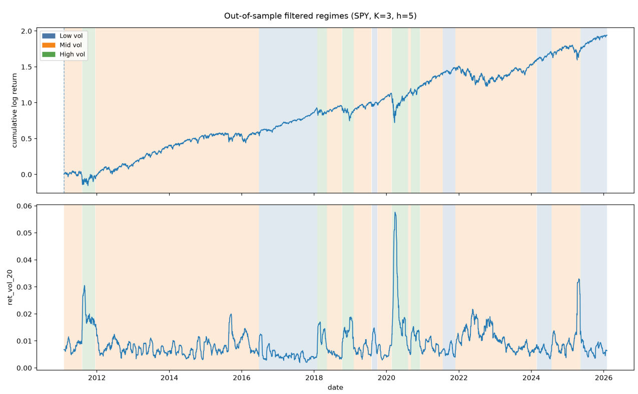

Figure 7: Out-of-sample filtered regimes for the K=3 model at h=5. Shading marks contiguous hard-state segments with opacity proportional to posterior confidence. Top: cumulative log return. Bottom: trailing realized volatility. The high-volatility state aligns with 2011, 2015–16, 2018, 2020 and 2022.

Table 13: What the inferred regimes do capture (K=3, h=5, out-of-sample). "Stress" is the top decile of realized volatility over the forecast window, so the unconditional rate is 10%. Lift is the ratio of the within-regime rate to that base rate. The high-volatility state occupies a sixth of the sample and contains nearly three-fifths of all stress events—the regimes are genuinely informative, just not incrementally so for the conditional mean.

Regime	% of time	Mean RV$^{(20)}$	P(stress)	Lift	% of all stress
Low vol	47.3	0.00623	0.021	0.21	9.8
Mid vol	36.4	0.00884	0.087	0.87	31.5
High vol	16.3	0.01707	0.360	3.60	58.7

6.14 Do the regimes mean anything?

It would be wrong to conclude that the HMM is failing to find structure. Figure 7 shows that the filtered high-volatility state aligns closely with recognized stress episodes, and it does so causally, without lookahead. The fits are also well-behaved: across all 15 folds no state ever collapses, the canonical volatility ordering is strictly monotone in every fold for $K \in \{2, 3\}$, and mean self-transition probabilities are 0.93, 0.81 and 0.68 for $K = 2, 3, 4$, so the inferred states are genuinely persistent rather than flickering.

Table 13 makes this quantitative, and the numbers are not marginal. The high-volatility state occupies 16% of the out-of-sample period but contains 59% of all volatility spikes: conditional on being in it, the probability of a top-decile spike over the next week is 36% against an unconditional 10%, a lift of 3.6×. As a stress *classifier* the causal HMM works well.

This is the sharpest form of the paper's result, and it admits a direct test. If the regime label is largely a re-encoding of trailing volatility, it should be recoverable from trailing volatility. Regressing the filtered hard state on the three HAR components alone—no other features, no HMM—recovers it with 95.7% accuracy against a 46.2% base rate. (This is an in-sample descriptive fit, not a forecast: the question is whether the regime variable is a function of information already present, not whether it can be predicted.)

That is the mechanism behind Section 6.6. The HMM state is a noisy, three-level, annually-refitted encoding of a quantity every model in the comparison already reads continuously and without estimation error. Adding it cannot supply information the model lacks, but it does supply estimation noise, which is why feature augmentation loses even with no partition to pay for. The gating mechanism then adds a second, larger cost on top.

The distinction matters for practice. Regime labels may still be valuable for communication, risk limits, or position sizing, and a 3.6× lift on stress is a usable signal for those purposes. Our result says only that they do not improve the conditional mean of realized volatility.

7 A Checklist

The audit that produced this paper reduces to a small number of checks that we would encourage as standard for regime-switching forecasting work:

(1) **Verify the target varies with the horizon.** Compare the marginal distributions of $y^{(h)}$ across h. If they are identical, the target is a single-period quantity at longer lead, not a multi-horizon forecast.
(2) **Use filtered, not smoothed, state probabilities**, and verify it numerically by checking that γ_t is unchanged when data after t is deleted.
(3) **Purge the training window by the horizon.**
(4) **Select hyperparameters inside the training window.** If the number of regimes is chosen on the test score, the test score is no longer an estimate of out-of-sample performance.
(5) **Give the baseline the same care as the model**—standardization, tuned regularization, and access to the same features. Include HAR and GARCH for volatility targets.
(6) **Run a gate-shuffle ablation**, and compare what correct timing buys against what the conditioning mechanism costs. Both quantities are measurable on the same scale, and only their difference is the result. Interpret the ablation across

many series: on a single asset it is too noisy to distinguish "no signal" from "signal too small to pay for itself".

(7) **Check that an ablation is not a duplicate.** Ours were: two of the four arms in our first design returned bitwise-identical numbers.
(8) **Relabel states canonically before pooling across folds**, or every per-regime table silently mixes different regimes together.
(9) **Check the regime variable is not redundant** with a feature the model already has. Regress the inferred state on your existing volatility features: ours was 95.7% recoverable, and nothing downstream could fix that.
(10) **Report persistence with overlapping-window emissions removed.** An HMM fitted to i.i.d. noise reports substantial regime persistence if its emission vector contains a rolling statistic. Refit without it and report both numbers.
(11) **Validate on a simulated world with known regimes.** If conditioning does not help where regimes provably exist, the architecture is at fault and no amount of real data will reveal it. If it does help there and not on real data, that is evidence about markets rather than about code.
(12) **Test the signal without the mechanism.** Before concluding that gating is what fails, feed the posteriors in as plain features. The two diagnoses imply different remedies.
(13) **Report the seed distribution**, not a single run.
(14) **Test significance** with a HAC-corrected Diebold–Mariano test and correct for the number of comparisons.
(15) **Report what your null excludes,** not just that you failed to reject. An equivalence bound distinguishes a measured null from an underpowered one.

8 Limitations

Our result is specific along several dimensions, and we do not want to overgeneralize from one asset.

Asset universe. We cover 20 liquid, exchange-traded instruments. They span asset classes but they are all liquid and all have long histories; genuinely illiquid or short-history assets, and instruments whose volatility is driven by scheduled events rather than by persistence, are not represented.

Expert families. We test ridge and gradient-boosted experts. The gating cost turns out to depend on the family—large and highly significant for ridge, indistinguishable from zero for boosting (Section 6.9)—so the decomposition in Section 6.6 describes linear experts and should not be assumed to generalise. What does generalise across both families is that the regime-conditioned model loses to a plain pooled ridge.

One family of regime detector. We use a Gaussian HMM on a three-dimensional emission vector, all of which is a function of past returns. A detector built on information genuinely outside the trailing-volatility span—order flow, options-implied measures, macroeconomic releases—would not be redundant in the way Section 6.6 shows ours is. We regard this as the most promising direction the paper points at, and the redundancy diagnostic of Section 6.6 as the check such work should report first.

Economic evaluation is narrow. We test volatility targeting and VaR on a single asset at a time. Regime information may matter more for problems our setup cannot see: cross-asset allocation, where correlations shift with regime [11], or drawdown control, where the timing of a regime call matters more than the accuracy of a conditional mean.

Gaussian HMM with a small emission vector. Richer emissions, non-Gaussian observation densities, or duration-dependent transitions could plausibly recover regimes that are less redundant with trailing volatility.

Conditional mean only. We evaluate point forecasts. Regime models produce predictive *distributions*, and their value may lie in tail quantiles or density forecasts, which our loss functions do not probe.

Fixed evaluation window. Results cover 2011–2026 out of sample. This period contains major stress episodes but is one realization of history.

Pairwise testing. We report Holm-corrected pairwise Diebold–Mariano tests against a fixed reference model. A Model Confidence Set [14] would characterize the set of models that cannot be distinguished from the best, which is a cleaner object when many models are compared at once; we use the pairwise form because our claims are specifically about the regime model against named benchmarks.

9 Conclusion

We set out to show that causally-inferred market regimes improve volatility forecasting, and found that they do not. Against properly specified benchmarks, with an embargoed split and hyperparameters selected inside the training window, an HMM regime-conditioned ridge is beaten by the same ridge fitted on pooled data in 58 of 60 asset-horizon pairs, and beats it significantly in none.

The more useful finding is why, and it is not the answer we first reached for. The regime signal is not absent: the inferred high-volatility state predicts stress with a 3.6× lift, and correct timing is worth a measurable 0.69 points of RMSE. Our initial reading was that the gating mechanism was simply too expensive, which predicts that using the posteriors as plain features should work. It does not: that variant still loses, by 1.16 points before any gating cost is paid. The regime state is 95.7% recoverable from features the model already has, so every route of injecting it adds estimation noise and no information, and gating then adds a further 1.59 points on top.

The distinction matters for what one should do next. "The mechanism is too expensive" invites better mechanisms. "The variable is redundant with what you already condition on" invites a different regime detector—one built on information outside the trailing-volatility span—or none at all. We would not have been able to tell these apart without testing the non-gating variants, and we would have reported the wrong recommendation.

The conclusion holds on every axis we pushed it. It survives nonlinear experts, a Model Confidence Set that considers all competitors at once, and a decision-relevant loss. The last of these is the only place the picture softens: under volatility targeting the regime models are statistically indistinguishable from the pooled ridge rather than behind it. We think that is worth saying plainly, both because it is true and because it shows the ranking under squared error is not automatically the ranking that matters. It does not change the recommendation, since the models that ignore regimes entirely are significantly ahead of both.

We were also able to recover a significant positive result by reintroducing four common evaluation shortcuts. Negative results in forecasting are hard to interpret in isolation, because a null can always be attributed to insufficient tuning. Showing the precise path from a null to a significant positive—and back—makes the failure mode concrete and, we hope, harder to reproduce by accident.

Reproducibility

All results are produced from raw data by three entry points—the study driver, the protocol ablation, and the seed sweep—and every table and figure in this paper is generated programmatically from the stored artifacts rather than transcribed. This is not incidental tidiness: the project this paper grew out of carried a hand-copied results table whose numbers matched no artifact in the repository, and automating the path from artifact to table is what caught it.

The correctness claims that matter are covered by tests, including the lookahead check described in Section 5: filtered probabilities are recomputed on truncated inputs and asserted equal, so a regression to smoothed posteriors would fail the suite rather than quietly improve the results. Code, configuration and artifacts accompany the submission.

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